

Swinburne University of Technology

COS80025 – Data Visualisation

Semester 2 - 2025

Final Project Report

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1. Introduction

1.1 Background and Motivation

One of the biggest issues affecting our world right now is climate change. Rising temperatures, unpredictable weather, melting ice caps, sea level rise, and an increase in the frequency of extreme weather events are all consequences of this transformation of human civilizations and natural surroundings, which is mostly caused by greenhouse gas emissions (Forster, Smith et al. 2023). It can be challenging to analyze large volumes of complex data without clear and useful visualization, which is necessary to make sense of these changes.

The topic was selected because climate change affects people all around the world and has significant economic and social ramifications in addition to being an environmental issue. According to research, rising temperatures have already made inequalities around the world worse and put more strain on agriculture and food systems (Diffenbaugh and Burke 2019, Yang, Tilman et al. 2024). Thus, my objective is to employ data visualization to assist show the relationship between environmental indicators, such temperature anomalies and carbon dioxide (CO₂) emissions, and more general social and economic results, like GDP and energy use.

The goal of this project is to use data from Our World in Data to visualize global CO₂ emissions from 2013 to 2023. Since emissions are expressed in millions of tonnes of carbon dioxide (MtCO₂), it is easy to see how various nations and areas have influenced global emissions throughout time. Users can compare nations, see how emissions have evolved, and pinpoint key contributors with interactive visualizations.

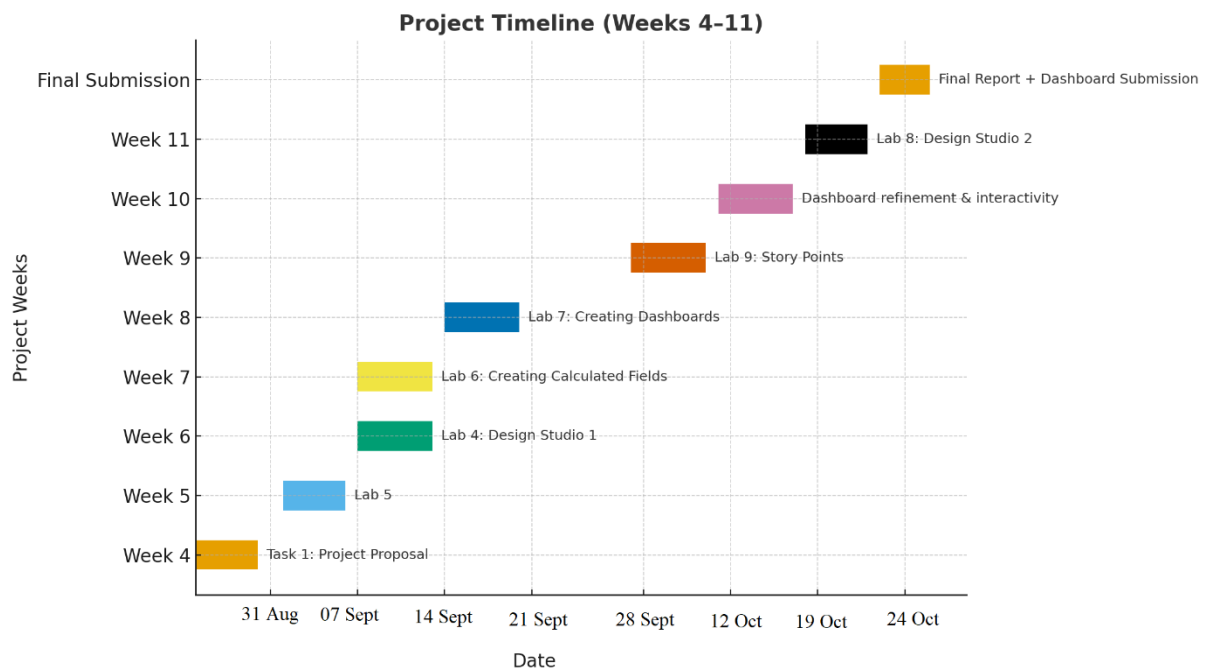
The visualisation is intended to help the general public better understand how the climate has evolved over the past ten years, as well as researchers and students studying climate trends and policymakers assessing the relationship between emissions and development. The ultimate goal of this project is to transform complicated environmental data into insightful, understandable information that promotes awareness, education, and well-informed decision-making.

1.2 Project Objectives

1. How have temperature anomalies, sea level rise, and global CO₂ emissions evolved from 2013 to 2023?
2. What regional or national variations in the effects of climate change are evident at this time?
3. How do environmental indicators and climate vulnerability connect to socioeconomic parameters like GDP?
4. In what ways might an interactive Tableau dashboard effectively convey and visualize these patterns to aid comprehension and decision-making?

1.3 Project Schedule

The below is the Project timeline along with the other lab submission since we will be learning and simultaneously implementing them.



2. Data

The dataset “**export_emissions**” used for the project consisting of 68 rows and 222 columns. It provides records of territorial CO₂ emissions for multiple countries and regions worldwide. Each column representing a country and each row corresponding to a specific year. CO₂ emission measured in million tonnes (MtCO₂) is also present

Variable Name	Description	Data Type
year	The year for Which permission data is recorded. Appears in the first column of the data set.	Temporal (integer)
country	The name of each country or region represented as individual columns for example Afghanistan, Australia, Brazil.	Categorical (Text)
co2_emissions_mtco2	The total annual territorial CO ₂ emissions for each country, measured in million tonnes (MtCO ₂)	Quantitative (Numeric)

2.1 Data Source

We used trustworthy, publicly accessible climate and socioeconomic data for the project. Primary sources consist of:

- Our World in Data – CO₂ and Greenhouse Gas Emissions(Ritchie, Rosado et al. 2023)
- NOAA – Climate Data Online(Online)
- NASA – Carbon Dioxide Vital Signs(Dioxide)
- Global Carbon Atlas(global and Atlas)
- Climate Watch – World Resources Institute(CLIMATEWATCH)

2.2 Data Processing

The data was standardized and cleaned before visualization were created. The following actions were performed on the data:

- **Deleting nulls values** to ensure the accuracy of graphs.
- **Removing duplicates** to prevent bias in the results.
- **Standardizing time frames** to ensure consistency (2013–2023).
- **Merging datasets** such as combining GDP per capita or population data with emissions for comparative analysis.

To clean and structure the proper required datasets we used use tools like:

- **Python, using its pandas and NumPy** libraries to transform the data as per requirement.
- **Excel** to perform manual checks.
- **Tableau** for integrating cleaned datasets and preparing them for visualization.

3. Requirements

3.1 Must have Features

The dashboard successfully delivers majority of the proposed must have features, specifically:

- The dashboard uses a number of visualizations to efficiently depict CO₂ emissions by nation and over time:
 - A global map of emissions by nation in 2023.
 - Global changes in CO₂ emissions from 2013 to 2023 are depicted in a time-series line picture.
 - The top ten countries that release CO₂ are shown in a bar chart.
 - A multi-country trend chart showing trends in emissions for a chosen group of countries.
- The dashboard satisfies the interactivity criteria by enabling users to filter by year and nation.
- Effective temporal coverage from 2013 to 2023 is attained, and the trends over that time are clearly visualized.

We were unable to include indications like temperature fluctuations, sea level rise, and the use of renewable energy sources in the finished dashboard, though, as suggested. The primary causes of this were scope constraints and data availability, as both factors need additional, consistent global datasets that were unavailable over the project's duration.

3.2 Optional Features

The final dashboard included a subset of the optional features:

- To provide a basic form of trend analysis and partial forecasting, a 3-year moving average trendline was added to the world emissions picture.
- Time and dataset constraints prevented the development of the other optional elements, such as correlation plots with socioeconomic variables and representations of extreme weather. It is determined that these need to be improved in the future.

4. Visualisation Design

Global CO₂ Emissions (2013–2023)

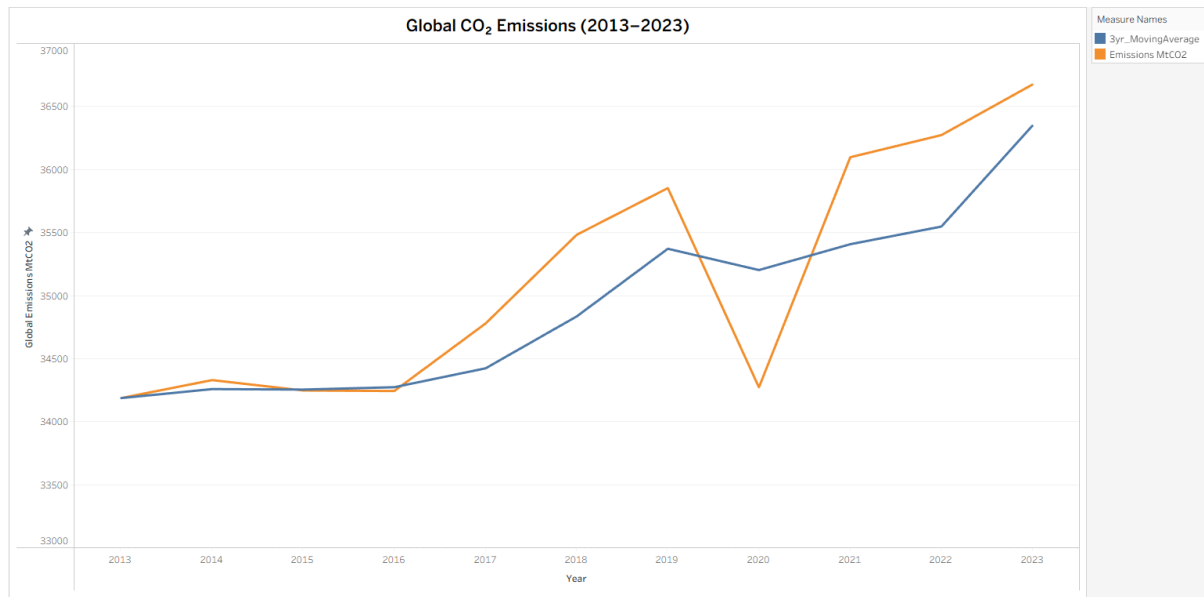


Figure 1. Global CO₂ Emissions (2013–2023)

The figure 1 chart presents the total global CO₂ emissions for each year from 2013 to 2023. The x-axis represents time in years and the y-axis shows total annual emissions in million tonnes (MtCO₂).

Global CO₂ emissions have been steadily rising for the last ten years, with only slight declines that most likely correspond to significant world events like the COVID-19 pandemic (2020). The general trend is still increasing despite short-term slowdowns.

This graph demonstrates how difficult it is still to cut emissions in spite of global accords like the 2015 Paris Accord. The minor decline in 2020 and subsequent recovery in 2021 emphasize how short-term emission reductions brought on by slowing economies are. It implies that in order to achieve long-term emission reductions, structural adjustments must be made.

Top 10 CO₂ Emitting Countries for year 2023.

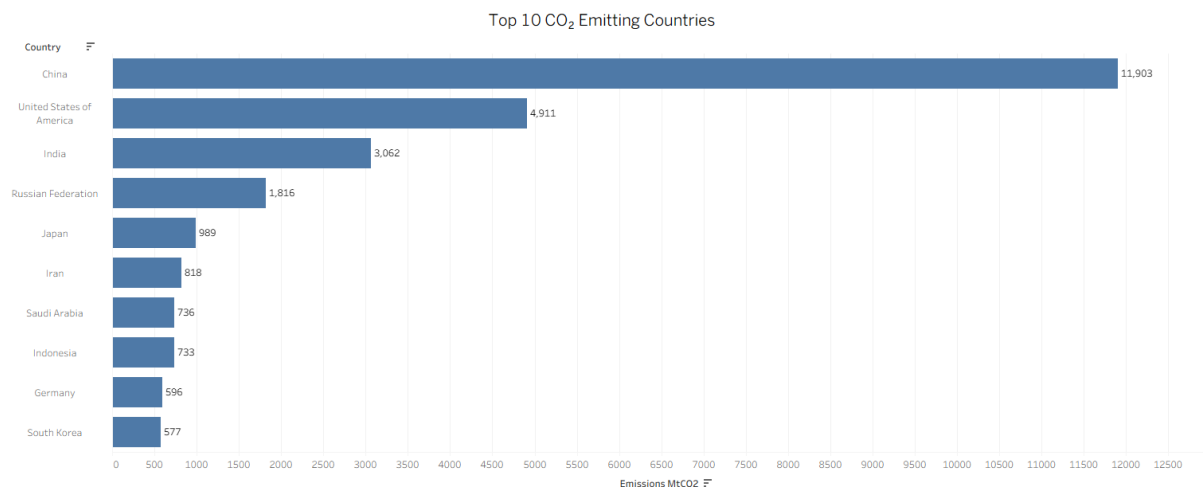


Figure 2. Top 10 CO₂ Emitting Countries

A horizontal bar chart ranking the 10 countries with the highest CO₂ emissions in the most recent year. Each bar represents the total emissions in MtCO₂, with colour used to denote regional grouping.

Together, the three leading countries China, United States of America, and India are accountable for over half of the world's CO₂ emissions. Most industrialized or fast expanding economies are the major polluters.

This visualisation underscores the unequal distribution of global emissions. It shows that industrial production and carbon emissions are closely related. Per-capita emissions, however, might vary greatly, which limits the usefulness of aggregate comparisons. Future iterations could provide a more equitable cross-country perspective by incorporating per-capita measures.

CO₂ Emissions by Country - 2023

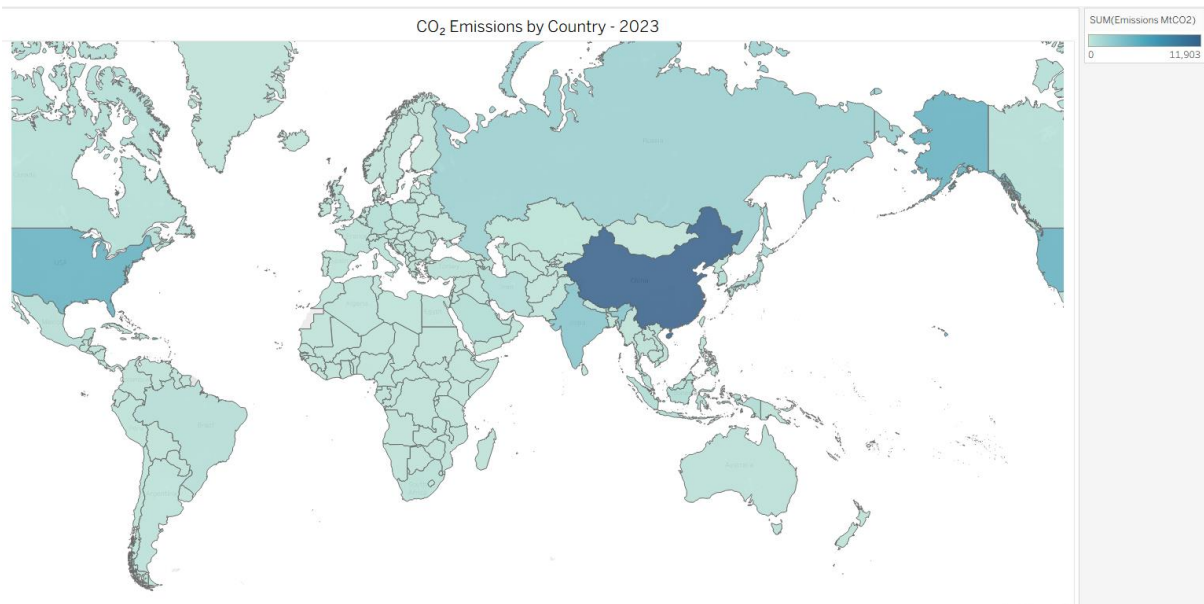


Figure 3. CO₂ Emissions by Country - 2023

A choropleth map displaying CO₂ emissions by country for a selected year 2023, using colour intensity to represent emissions levels (darker shade = higher emissions).

High emission intensities are concentrated in Asia, North America and parts of Europe while Africa and small island nations contribute minimally to global total emissions.

The map instantly conveys a sense of spatial inequality. It emphasizes how areas most impacted by climate change like Africa and Oceania contribute the least to the issue. Nevertheless, this map is restricted by absolute values; a fairer comparison might be obtained by normalizing emissions per GDP or per million. However, by using visual geography, it successfully attracts non-technical audiences.

Emissions Trends - Selected Countries (2013–2023)

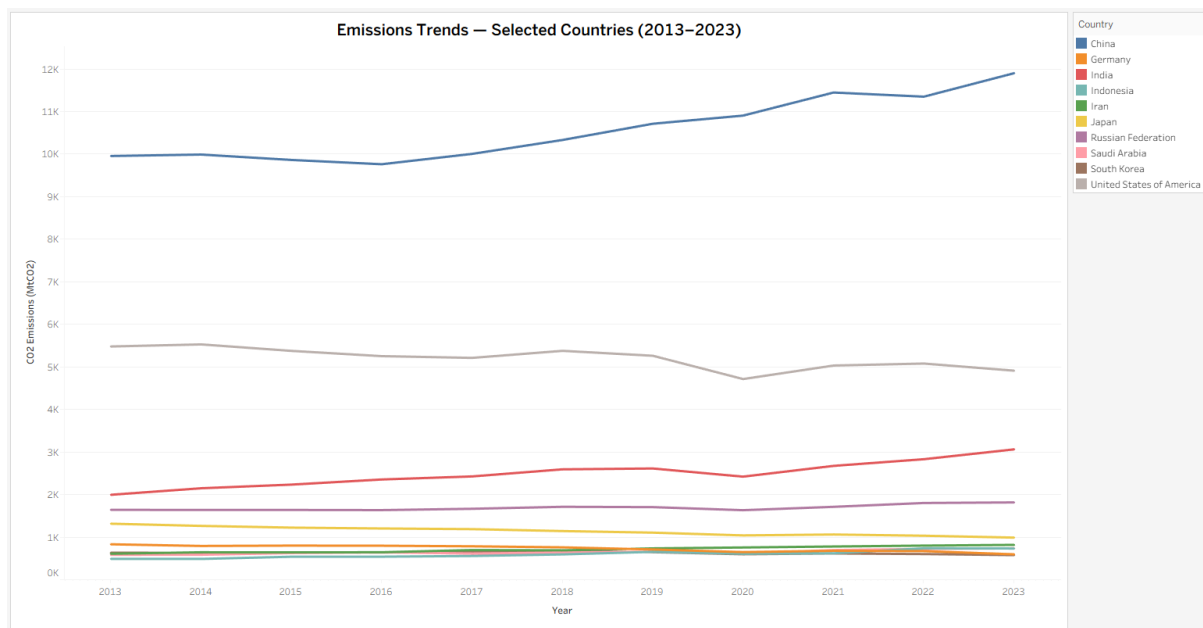


Figure 4. Emissions Trends - Selected Countries (2013–2023)

A multi-line chart showing annual CO₂ emissions for selected countries over the 2013 to 2023. Each line represents a different country.

While emissions from developed nations like the US, Germany, and the UK are consistent or dropping over time, those from growing economies like China and India show steady growth.

This comparison shows how emissions are changing globally. Developing economies continue to increase their carbon output as they industrialize but developed nations have started to decouple economic growth and emissions. It shows the need for distinct climate strategies under global frameworks as well as economic advancement. The graph works well for comparing time periods, but if too many nations are shown, it may get cluttered. Thorough screening is necessary.

CO₂ Emissions by Countries from 2013- 2023

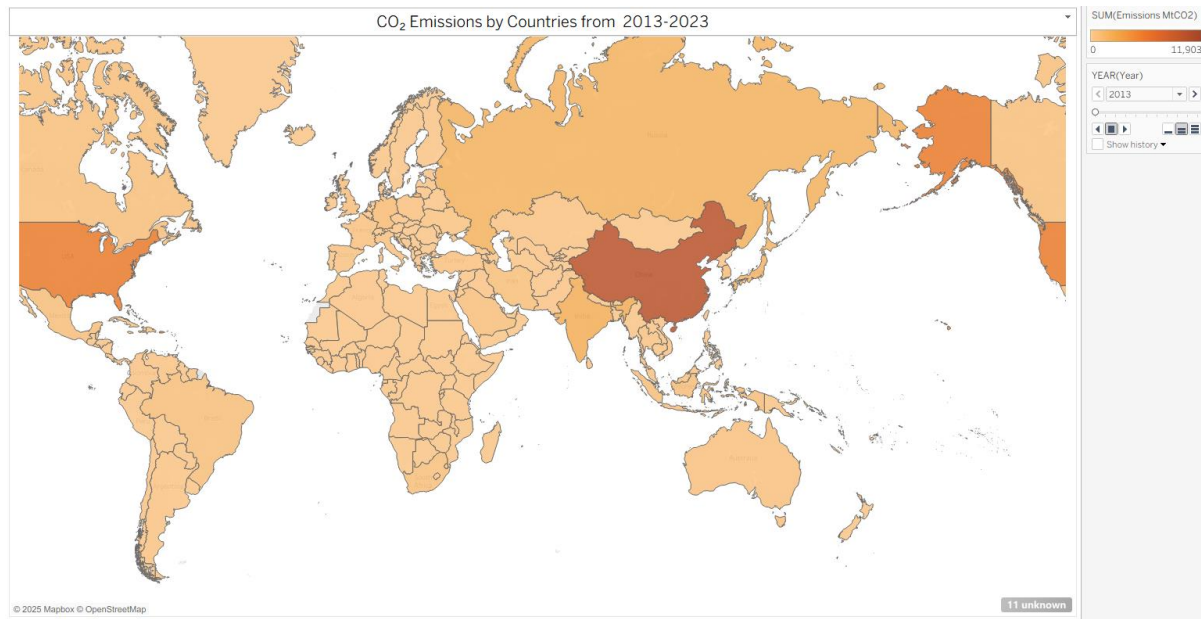


Figure 5. CO₂ Emissions by Countries from 2013- 2023

This animated choropleth map illustrates the global changes in carbon dioxide (CO₂) emissions between 2013 and 2023. Countries are colored according to the amount of CO₂ emissions they produce, measured in million tonnes (MtCO₂), with each frame of the animation representing a year. Higher emission levels are indicated by darker hues. Users may see how emissions change on a global and regional scale as the map dynamically refreshes over time.

The animation shows that emissions have clearly and steadily increased over the past ten years in most regions, with Asia seeing particularly rapid development and North America and Europe continuing to have high levels. Despite international mitigation efforts, global emissions are still rising overall, even if certain countries have seen slight decreases.

Users can see slow but significant changes in CO₂ emissions because to the dynamic map's good integration of temporal and spatial dimensions. The gradual darkening of colours in developing nations emphasizes the continued industrialization and rising energy use.

The motion makes sense to non-technical viewers right away since it emphasizes the cumulative aspect of emissions rise from a communication standpoint. But using absolute emission figures can hide per-capita or efficiency disparities while visually exaggerating the impact of large nations. Future versions might incorporate an intensity scale or per-capita mode option to offer a more fair comparison.

5. Validation

This section explains how the interactive Tableau dashboard was put into use and assesses how well it conveys information regarding CO₂ emissions worldwide from 2013 to 2023. With the help of filtering, highlighting, and temporal animation, users may dynamically examine data on the dashboard, which integrates all of the visualisations created in the visualisation design phase.

5.1 Dashboard Overview

The final dashboard integrates the following components:

1. **Global CO₂ Trend (Line Chart)** shows all the overall emission pattern overtime.
2. **Top 10 emitters (Bar Chart)** displays the highest everything countries for selected year 2023.
3. **World map (Choropleth)** provide special context of emission distribution.
4. **Country Trends (Multi-Line)** compares historical trends for selected countries.

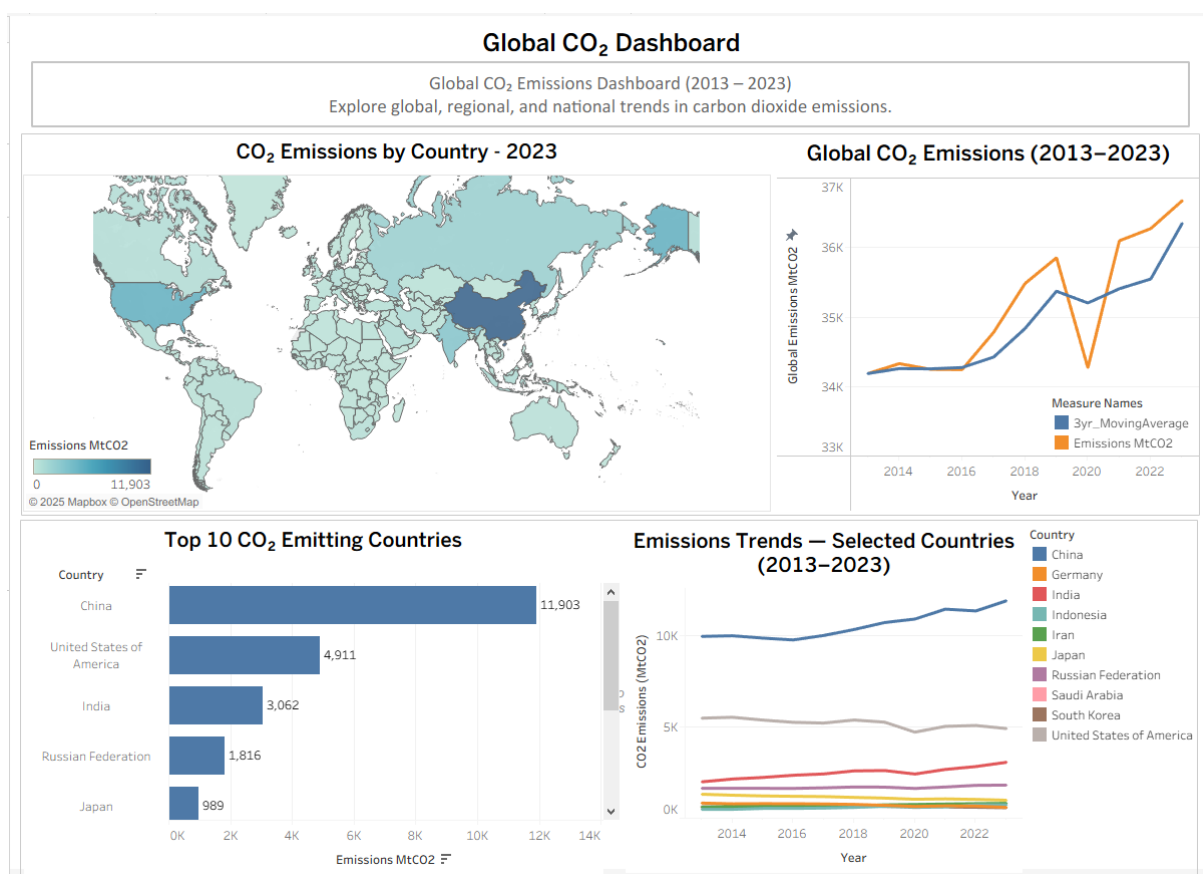


Figure 6. Global CO₂ Dashboard

5.2 Interactivity and User Features

Features	Functionality
Year Filter (Slider)	Allows users to select and view a specific year or animate across all years
Country Filter (Dropdown)	Enables comparison of one or more countries across all charts.
Map Selection Filter	Clicking a country on the map filters all other views automatically.
Hover Tooltips	Display detailed information such as CO ₂ value, 3 year average, GDP per capita etc. when hovered over a mark.
Animated Timeline	Shows progressive emission increases from 2013-2023.
Highlight Actions	Hovering over a country or region emphasises related data across charts.

6. Conclusion

Through the visualization and interpretation of global CO₂ emissions from 2013 to 2023, this project aimed to convert complicated environmental statistics into understandable, interactive insights. The dashboard skilfully blends spatial, temporal, and comparative studies using Tableau to show how emissions have changed from one country or region to another over the last ten years.

The graphics show that CO₂ emissions are still increasing globally in spite of international climate agreements and awareness campaigns. While developed economies like the US, UK, and Germany show stabilizing or declining patterns, developing countries like China and India show tremendous development as industrialization spreads. The animated map in particular highlights the ongoing difficulty of lowering greenhouse emissions at scale and offers a powerful visual story of this slow but constant global growth.

Users can examine the extent and distribution of emissions by utilizing the dashboard's integration of several global, regional, and country viewpoints. It encourages a better comprehension of the ways in which geography, industrial activity, and economic development affect environmental results.

Even though data constraints prevented the inclusion of several indicators, such temperature anomalies and the use of renewable energy, the project effectively illustrates how visualisation may increase the accessibility and significance of climate data. The final dashboard serves as a tool for researchers, politicians, and the general public to interact with the realities of climate change through data-driven storytelling, not only by showcasing trends but also by promoting awareness.

References

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Appendix

Appendix A1: Data preparation and processing

I use the below Python code snippet to clean and format the data set before importing it into Tableau.

```
import pandas as pd

df = pd.read_excel("export_emissions.xlsx")

df = df.dropna().drop_duplicates()

df = df[(df['year'] >= 2013) & (df['year'] <= 2023)]

df_long = df.melt(id_vars=['year'], var_name='country',
value_name='co2_emissions_mtco2')

df_long.to_csv("cleaned_export_emissions.csv", index=False)
```

Appendix A2: Manual Data Checks

The “export_emissions” sheets containing the raw data set along with the empty columns and null values as shown below in figure 7.

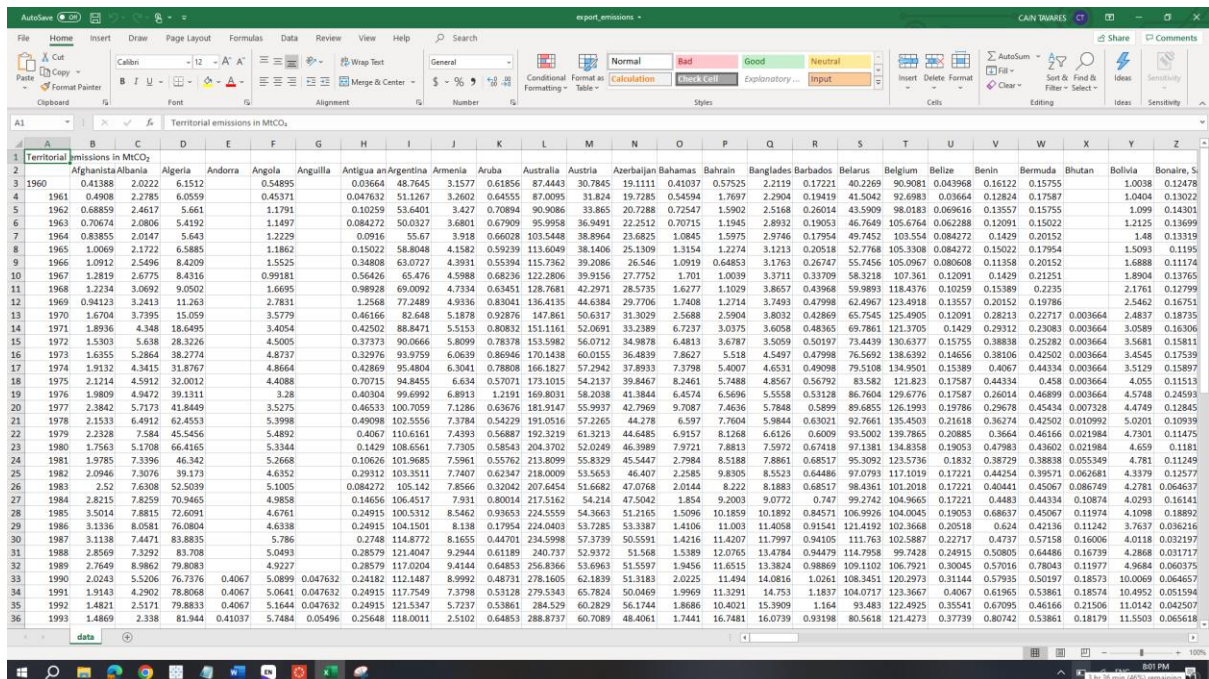


Figure 7. Raw Dataset

Appendix A3: Clean Dataset

The “Cleaned_data_emissions.csv” contains the cleaned data extracted using the pandas python library and we selected the below 3 columns i.e. year, country and Emissions_MtCO₂ was imported to Tableau Desktop as shown in the below figure 8.

	A	B	C	D	E	F	G
1	Year	Country	Emissions_MtCO ₂				
2	2013	Afghanistan	9.2285				
3	2014	Afghanistan	9.086				
4	2015	Afghanistan	9.6702				
5	2016	Afghanistan	8.9064				
6	2017	Afghanistan	9.6774				
7	2018	Afghanistan	10.6019				
8	2019	Afghanistan	10.8251				
9	2020	Afghanistan	11.6059				
10	2021	Afghanistan	10.2719				
11	2022	Afghanistan	10.5578				
12	2023	Afghanistan	11.0202				
13	2013	Albania	5.2875				
14	2014	Albania	5.9997				
15	2015	Albania	4.7121				
16	2016	Albania	4.632				
17	2017	Albania	5.293				
18	2018	Albania	4.895				
19	2019	Albania	4.8269				
20	2020	Albania	4.7111				
21	2021	Albania	5.1341				
22	2022	Albania	5.1734				
23	2023	Albania	5.1443				
24	2013	Algeria	140.7864				
25	2014	Algeria	151.2828				
26	2015	Algeria	160.0875				
27	2016	Algeria	158.4816				

Figure 8. Cleaned Dataset

Appendix A4: Calculated Fields

Calculated Field	Description	Purpose
3yr_MovingAverage	WINDOW_AVG(SUM([Emissions MtCO ₂]), -2, 0)	Smooths short-term fluctuations in emission trends.
Top10	Top 10 emission causing countries consisting of China, Germany, India, Indonesia, Iran, Japan, Russian Federation, Saudi Arabia, South Korea, United States of America.	Highlighting the top 10 CO ₂ emitting countries for focused comparison in bar charts and dashboards.